Curriculum Vitæ Bhavin K. Moriya

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RESEARCH INTERESTS

Mathematical modelling and applications, statistics, machine learning, software development, and privacy-enhancing technologies (homomorphic encryption, privacy-preserving analytics).

PREVIOUS POSITION

A **Data Scientist / Research Assistant** at **Hochschule Esslingen, Germany** during **Nov 2023 – Dec 2025**, working on the **AnoMoB project** (BMBF-funded research on anonymised capture and use of mobility data).

- **AnoMoB Project:** Developed first-of-its-kind encrypted comparison operations and non-linear function evaluations using **OpenFHE (C++)** and **TFHE-rs (Rust)**.
- **Performance Benchmarking:** Co-authored research comparing runtime and memory consumption of the **CKKS and TFHE** schemes, published in the Cryptology ePrint Archive (2025).
- **Data Engineering:** Built production-quality software tools for statistical analysis of mobility data using **Python, polars, and geopandas**.
- **Sustainability Analysis:** Leveraged **SUMO** to simulate movement data and conducted comparative analyses of travel modalities focusing on **CO2 emissions** and optimal route duration.
- **Decision Support for Sustainable Mobility:** Developed a **decision-support tool** for the AnoMoB project using a weighted scoring system to evaluate trade-offs between travel time and CO2 emissions, based on Dockerized routing engines (ORS/OTP).

PROFESSIONAL EXPERIENCE

- **Wissenschaftlicher Mitarbeiter** — Hochschule Esslingen, Esslingen am Neckar, Germany — **Nov 2023 to Dec 2025**.
- **Assistant Professor** — Universidade Federal de Viçosa, Viçosa, Brazil — **Nov 2015 to Nov 2023**.
  - Taught undergraduate and master's-level mathematics; supervised 3 dissertations.
  - Led a FAPEMIG-funded international research project on zero-sum problems over finite abelian groups.
  - Extensive collaboration with international researchers, resulting in 16+ peer-reviewed publications.
- **Post-Doctoral Fellow** — University of Brasilia, Brasilia, Brazil — **Nov 2014 to Nov 2015**.

- **Visiting Fellow** — Chennai Mathematical Institute (CMI), Chennai — **Sep 2014 to Nov 2014**.
- **National Academy of Sciences Research Associate** — Harish Chandra Research Institute, Allahabad — **Aug 2013 to Aug 2014**.
- **Post-Doctoral Fellow** — The Institute of Mathematical Sciences, Chennai — **Aug 2011 to Aug 2013**.

Skills

Programming: Python (proficient), C++, Rust, SQL, LaTeX, Go, Julia
Data & Simulation: polars, pandas, geopandas, SUMO, scikit-learn, Docker, Streamlit
Workflow: Git, Agile Research, LLM-assisted development
Languages: English (fluent), German (A2, working towards B1), Portuguese (fluent), Hindi (native), Gujarati (native)

- Proficient in working with homomorphic encryption. In particular, have used OpenFHE and TFHE-rs to implement custom functions — for example comparison functions, polynomial evaluation, square roots, etc.
- Proficient in machine learning techniques, including neural networks, recurrent neural networks, bagging and boosting models, regression, and classification.
- Working knowledge of finance: derivative pricing (Black–Scholes–Merton), portfolio optimisation (CAPM, Markowitz), Monte Carlo simulation, and credit-risk modelling (PD/LGD/CCF).

Selected Projects

- **FEM Solver — Finite Element Method from Scratch**
- **AudioMNIST** — A spoken digit recognition system.
- **Audio Digital Signal Processing** — An interactive web application exploring the mathematics and applications of DSP for audio. Demo
- **SigVision** — Signal processing for computer vision. Demo
- **Credit Risk Model — PD · LGD · CCF**
- **Vascular Surrogate Models** — Accelerating intracranial aneurysm CFD.
- **More Projects**

Education

- Ph.D. in Mathematics, Harish Chandra Research Institute, Allahabad, 2011.
  Thesis: Some Zero Sum Problems in Combinatorial Number Theory
- M.Sc. in Mathematics, The Maharaja Sayajirao University of Baroda, Vadodara, 2005.
- B.Sc. (Major: Mathematics), The Maharaja Sayajirao University of Baroda, Vadodara, 2003.

PUBLICATIONS

1. (with C. Krüger & D. Schoop) A Performance Comparison of the Homomorphic Encryption Schemes CKKS and TFHE, Cryptology ePrint Archive, Paper 2025/1460, 2025.
2. Signed zero sum problems for metacyclic groups. *Periodica Mathematica Hungarica*, pages 1–16, (2024).
3. (with FEB Martínez, A Lemos, S Ribas) The main zero-sum constants over $D_{2n} \times C_2$. *SIAM Journal on Discrete Mathematics*, 37 (3), pages 1496–1508, (2023).
4. (with A Lemos, AO Moura, AT Silva) On the Number of Weighted Zero-sum Subsequences. *Periodica Mathematica Hungarica* 87, pages 366–373, (2023).
5. (with A Lemos, BK Moriya, AO Moura, AT Silva) A Variant of Harborth Constant. arXiv preprint arXiv:2209.14784.
6. (with A Lemos, A de Oliveira Moura, AT Silva) On the number of fully weighted zero-sum subsequences. *International Journal of Number Theory* 15 (05), (2019), 1051–1057.
7. (with A Cambraia Jr, MP Knapp, A Lemos, PHA Rodrigues) On prime factors of Mersenne numbers. *Palestine Journal of Mathematics* 11 (2) (2022), 449–456.
8. (with S D Adhikari, Eshita Mazumdar) Relation between two weighted zero-sum constants. *INTEGERS* 16, (2016), 1–13.
9. (with S Gun) Ramanujan, Quasi-Modular Forms and Transcendence. *Submitted.*
10. On weighted zero sum subsequences of short length. *INTEGERS* 14, A21 (2014), 1–8.
11. (with C J Smyth) Index-Dependent divisors of coefficients of modular forms. *Int. J. Number Theory* 9 (7) (2013), 1841–1853.
12. (with M N Chintamani, W D Gao, P Paul, R Thangadurai) New upper bounds for the Davenport and for the Erdős-Ginzburg-Ziv constants. *Arch. Math. (Basel)* 98 (2012), no. 2, 133–142.
13. (with M N Chintamani) Generalizations of some zero sum theorems. *Proc. Indian Acad. Sci. Math. Sci.* 122 (2012), no. 1, 15–21.
14. (with S D Adhikari, M N Chintamani, Geeta) The Cauchy-Davenport theorem: various proofs and some early generalizations. *Math. Student* 79 (2010), no. 1–4, 109–116.
15. On zero sum subsequences of restricted size. *Proc. Indian Acad. Sci. (Math. Sci.)* 120 (2010), no. 4, 395–402.
16. (with M N Chintamani and P Paul) The number of weighted n-sums. *Int. J. Mod. Math.* 5 (2010), no. 2, 215–222.
17. (with S D Adhikari, M N Chintamani and P Paul) Weighted sums in finite abelian groups. *Unif. Distrib. Theory* 3 (2008), no. 1, 105–110.

Talks and Colloquiums

1. Delivered invited talks on Ramsey-type theorems at the University of Viçosa in August 2017 and February 2016, and at the University of Brasilia in November 2015; invited talk on zero-sum problems at the University of Brasilia in May 2015.
2. Conducted tutorials for the course on p-adic numbers at IMSc, Chennai, in 2013.
3. Presented Hensel's Lemma at IMSc, Chennai, on April 16, 2013.
4. Delivered lectures and tutorials during the Summer Program at IMSc, Chennai, May 25 – June 20, 2012.
5. Invited talk at Séminaire CAESAR de combinatoire additive, Université Pierre et Marie Curie (Paris 6), May 31, 2012: "Zero-sum subsequences of short length with weights."
6. Mathematics colloquium at the Institute of Mathematical Sciences, Chennai, November 24, 2011: "Some zero sum problems in combinatorial number theory."
7. Delivered a series of six lectures on number theory during the Summer Program in Mathematics (SPIM) 2011 at Harish Chandra Research Institute.

Academic Service

1. Organized the Summer School at the University of Viçosa, Brazil, October 2021 – September 2022.
2. Participated in the committee for the Simpósio de Integração Acadêmica — SIA 2016.
3. Organized the Summer School at the University of Viçosa, Brazil, January 9 – February 22, 2017.
4. Conducted tutorials at SPIM 2014, AIS Combinatorics 2014, and the Annual Foundation School — Part I (2013, Kerala School of Mathematics).
5. Conducted tutorials for courses on analytic number theory (KIIT University, 2013), algebra (2013), and p-adic numbers (2013).
6. Co-organized a summer program for M.Sc. students at IMSc with Dr. S. Gun, May 25 – June 20, 2012.

Supervision

- **Master's theses** (Universidade Federal de Viçosa): Cynthia Martins Diorio, *Corpos Finitos e Polinômios de Permutação* (2019); Luana Souza Marques, *Método Isoperimétrico em Teoria Aditiva dos Números* (2018).
- **PICME graduate**: Luana Costa, *Aplicação dos Corpos Finitos* (2019).

Projects and Grants

- **Zero Sum Problems over Finite Abelian Groups** — FAPEMIG-funded research project (team member with A Lemos Cardoso Júnior, AO Moura, AT da Silva, S Ribas, FEB Martínez).
- Awarded an ICM Grant by the National Board of Higher Mathematics (NBHM) to attend the International Congress of Mathematicians (ICM 2014), Seoul, South Korea.

- NBHM Travel Grant for the CIMPA-ICTP Research School on Algebraic Curves over Finite Fields, Manila, Philippines, 2013.
- NBHM scholarship for Ph.D. studies (2005); Dr. M. S. Patel Post Graduate Scholarship (2006).
- Grants for schools and conferences: CANT 2012 (CIRM Marseille), Codes over Rings (METU, 2008), ICM 2010 (Hyderabad).

TEACHING

- Assistant Professor (Universidade Federal de Viçosa, 2015–2023): 29 undergraduate and graduate courses including Calculus I & II, Linear Algebra, Algebra I & II, Rings and Modules, Finite Mathematics, and Special Topics in Mathematics; organized and taught a course in Analysis at the 2017 Summer School.
- Course on Additive Number Theory, University of Brasilia, August – December 2015.
- Tutorials and lecture series at IMSc, HRI, Kerala School of Mathematics, and KIIT University (2011–2014).

CONFERENCES AND WORKSHOPS (SELECTED)

1. Conference on Anonymization of Integrated and Georeferenced Data, Berlin, October 7–8, 2024.
2. 24th Privacy Enhancing Technologies Symposium (PETS), University of Bristol, UK, July 15–20, 2024.
3. 12th Privacy Enhancing Techniques Convention (PET-CON 2023), TU Berlin, March 13–14, 2024.
4. AnoSiDat Kongress: Anonymisierung für eine sichere Datennutzung, Lübeck, April 16–17, 2024.
5. International Conference in Number Theory and Physics, IMPA, Rio de Janeiro, June 2015.
6. International Congress of Mathematicians 2014, Seoul (ICM grantee).
7. Advanced Instructional School in Algebraic and Analytic Number Theory, HRI, December 2007; International Conference on Number Theory and Cryptography, HRI, 2007.

REFERENCES

- Prof. Dr. Dominik Schoop
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- Prof. Jean-Paul Allouche
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